PYTHON FUNDAMENTALS

ASSIGNMENT 02

Logic Building Practice
Conditionals · Loops · Functions · Algorithms

YOUR LEARNING ROADMAP

01 · Decision Making Use if/elif/else to choose actions.	**02 · Loops** Repeat tasks and process number ranges.
03 · Functions Break problems into reusable pieces.	**04 · Problem Solving** Build logic with digits, primes, and games.

Mindset: The goal is to build programming logic. Try writing the code yourself before looking at hints. It is okay to work out the mathematical idea separately—as long as you implement the Python solution yourself. Test your programs with different inputs.

PART A · ORIGINAL ASSIGNMENT

QUESTION 01 · Salary Tax Rate

Ask the user to enter a salary. Use conditional statements to determine the tax rate:
• Salary below 30,000 → 5%
• Salary from 30,000 to 70,000 → 15%
• Salary above 70,000 → 25%.

Practice focus: if / elif / else, comparisons

QUESTION 02 · Even Numbers in a Range

Write a function that takes two integers, `a` and `b`, and prints every even number between them, inclusive.

Practice focus: functions, loops, modulo (%)

QUESTION 03 · Print the Digits

Write a function that prints the digits of an integer one by one. For example, for `n = 312`, the digits are 3, 1, and 2.
Hint: The rightmost digit is `n % 10`. Integer division with `n // 10` removes the rightmost digit.

Practice focus: while loops, //, %

QUESTION 04 · Count the Digits

Write a function that returns the number of digits in an integer.

Practice focus: loops, counters, integer division

Hint: Consider what your function should return for 0 and how to handle negative numbers.

QUESTION 05 · Sum of Digits

Write a function that returns the sum of the digits of an integer. For example, the sum of the digits of 312 is 6.

Practice focus: loops, modulo (%), integer division (//)

PART A · ORIGINAL ASSIGNMENT (CONTINUED)

QUESTION 06 · Divisible by Both 3 and 5

Print all numbers from 1 to 100 that are divisible by both 3 and 5.

Practice focus: for loops, logical conditions, modulo (%)

QUESTION 07 · Positive or Negative Until Quit

Continuously ask the user to enter a number. Print whether it is positive or negative, and keep going until the user enters `Quit`.

Practice focus: while loops, strings, conditionals

Hint: Check for the quit command before converting the input to a number. Decide how to handle zero.

QUESTION 08 · Function-Based Calculator

Create a function `calculator(a, b, operation)` that performs addition, subtraction, multiplication, or division depending on whether `operation` is `+`, `-`, `*`, or `/`.

Practice focus: functions, if/elif/else, division-by-zero handling

QUESTION 09 · Prime Number Checker

Write a function `is_prime(n)` that returns `True` if n is prime and `False` otherwise.

Practice focus: loops, Boolean values, divisibility

Hint: 2 is the smallest prime. Numbers below 2 are not prime. Check possible divisors from 2 up to (but not including) n; later, try improving the range.

QUESTION 10 · Number Guessing Game

Choose a secret number in your code and ask the user to guess it. Print `Too high` if the guess is greater, `Too low` if it is smaller, and `Correct!` if it matches.

Practice focus: conditionals, user input, comparisons

PART B · EXTRA PRACTICE

Ten additional challenges to strengthen your logic-building skills. Start with a simple solution, then improve it.

QUESTION 11 · Grade Calculator

Ask for a score from 0 to 100 and print a grade using your own clearly stated ranges. Also reject scores outside the valid range.

Practice focus: conditionals, comparisons, validation

QUESTION 12 · Multiplication Table

Ask for an integer and print its multiplication table from 1 to 10.

Practice focus: for loop, range(), formatted output

QUESTION 13 · Factorial Function

Write a function factorial(n) that returns n! for a non-negative integer. Test it with 0 and 5.

Practice focus: functions, loops, accumulator

QUESTION 14 · Find the Largest of Three

Ask for three numbers and print the largest without using max().

Practice focus: if / elif / else, comparisons

QUESTION 15 · Count Vowels

Write a function that takes a string and returns how many vowels (a, e, i, o, u) it contains, ignoring letter case.

Practice focus: strings, loops, membership tests

QUESTION 16 · Reverse a Number

Write a function that reverses the digits of a positive integer. For example, 1234 becomes 4321.

Practice focus: while loops, //, %, arithmetic

QUESTION 17 · Fibonacci Sequence

Ask how many terms to display, then print that many terms of the Fibonacci sequence, starting with 0 and 1.

Practice focus: loops, variable updates

QUESTION 18 · Armstrong Number

Write a function that checks whether a non-negative integer is an Armstrong number. For example, 153 is an Armstrong number because $1^3 + 5^3 + 3^3 = 153$.

Practice focus: digit extraction, loops, exponentiation

QUESTION 19 · Password Attempts

Set a practice password in your program. Allow up to three attempts, then print whether access is granted or the user is locked out.

Practice focus: while loop, counters, conditionals

QUESTION 20 · Menu-Driven Calculator

Create a calculator that repeatedly displays an operation menu, performs the selected calculation, and stops when the user chooses Exit.

Practice focus: functions, loops, conditionals, input validation

SUBMISSION & SELF-CHECK

Before committing to GitHub

- ■ Use meaningful function and variable names.
- ■ Test boundary cases such as 0, negative numbers, and invalid input where relevant.
- ■ Use `//` when removing digits from integers; avoid accidentally turning them into floats.
- ■ Keep each solution readable and add comments only where they clarify the logic.
- ■ Add a README describing the concepts practised and how to run the solutions.

Learning note: This worksheet provides problem statements and selected hints, not complete solutions. Solve independently and commit your code when you understand it.